



**United International University**  
**Department of CSE**

**Course Code:** CSE 4326

**Course Name:** Microprocessors and Microcontrollers  
Laboratory

**Experiment no** : 02

**Experiment Name:** Interfacing MQ-2 Gas Sensor with  
Arduino and OLED Display

**Submitted by (Group no.):** 01

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**Section:** A

**Date of Performance:** 18 July 2026

**Date of Submission** : 25 July 2026

## **Objective:**

The objective of this experiment is to learn how to interface an MQ-2 gas sensor with Arduino Uno and display the measured gas value on an OLED display. This experiment also demonstrates how Arduino can respond automatically when gas is detected by activating different output devices such as an LED and a buzzer. Through these tasks, I learned how to read sensor data, display information on an OLED screen, and build a simple gas detection and warning system.

## **Apparatus:**

- Arduino Uno
- MQ-2 Gas Sensor
- OLED Display
- Buzzer
- LED
- Breadboard
- Jumper Wires
- Resistors
- Arduino IDE

## **Theory:**

The MQ-2 gas sensor is used to detect gases such as LPG, methane, propane, and smoke. It produces an analog output that changes according to the amount of gas present in the surrounding environment. Arduino reads this value and performs different actions based on the programmed conditions.

The OLED display is used to show the gas sensor reading and warning messages. It communicates with Arduino through the I2C interface, making it simple to connect while providing a clear display.

A buzzer is an output device that produces sound when it receives an electrical signal. It is commonly used as an alarm to notify users when gas is detected.

An LED is a simple visual indicator. It can be turned on or off by Arduino depending on the gas sensor reading. In this experiment, the LED indicates the presence of gas when the measured value exceeds the predefined threshold.

By combining these components, a simple gas monitoring and warning system can be developed. The system provides both visual and audible indications whenever gas is detected.

## Solution of the mentioned problems:

### Problem 1: Display the Gas Sensor Data on the OLED

#### Solution:

In this task, I connected the MQ-2 gas sensor and the OLED display with Arduino Uno. The sensor continuously measured the gas concentration and sent the analog value to Arduino. After reading the sensor value, the program displayed the measured gas data on the OLED screen in a readable format.

The circuit was tested several times by exposing the sensor to different conditions. I observed that the displayed value changed according to the amount of gas detected by the sensor, which confirmed that both the sensor and the OLED display were working correctly.

#### Code for Problem 1:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define SCREEN_ADDRESS 0x3C
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
const int sensorPin = A0;
#define RL 10
#define m -0.263
#define b 0.42
#define Ro 20

void setup()
{
  Serial.begin(9600);
  Serial.println("MQ2 Warming Up...");
  delay(20000);
  if (!display.begin(SSD1306_SWITCHCAPVCC, SCREEN_ADDRESS))
  {
    Serial.println("OLED Failed");
    while (1);
  }
  display.clearDisplay();
  display.setTextColor(WHITE);
}

void loop()
```

```

{
  float sensorValue = analogRead(sensorPin);

  float VRL = sensorValue * (5.0 / 1023.0);
  float Rs = ((5.0 * RL) / VRL) - RL;
  float ratio = Rs / Ro;
  float ppm = pow(10, ((log10(ratio) - b) / m));
  Serial.print("ADC: ");
  Serial.print(sensorValue);
  Serial.print(" PPM: ");
  Serial.println(ppm);
  display.clearDisplay();
  display.setTextSize(2);
  display.setCursor(0, 0);
  display.println("MQ-2");
  display.setTextSize(1);
  display.setCursor(0, 30);
  display.print("ADC : ");
  display.println(sensorValue);
  display.setCursor(0, 45);
  display.print("PPM : ");
  display.print(ppm, 1);
  display.display();
  delay(500);
}

```

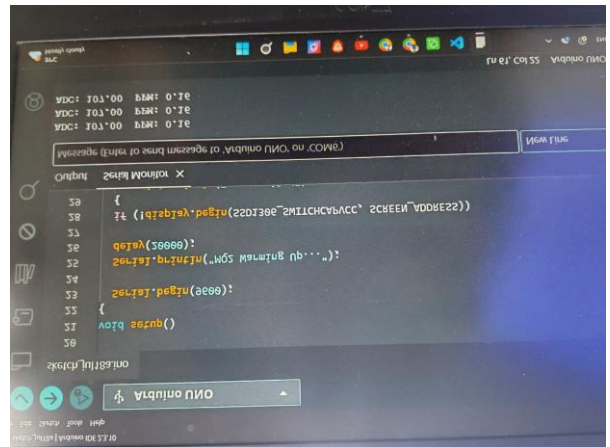
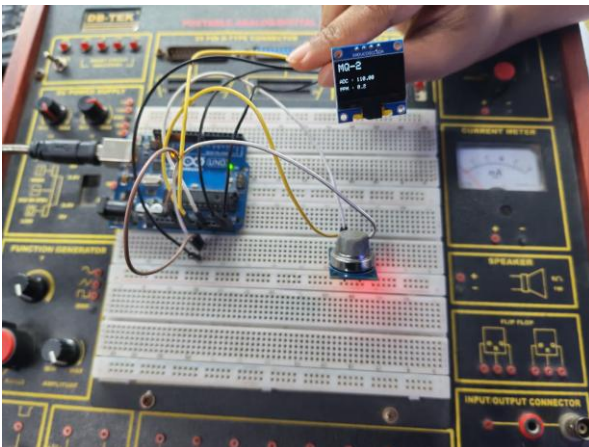


Figure: Problem 1

## Problem 2: Turn ON a Buzzer When the Gas Value Exceeds the Threshold

### Solution:

In this task, I connected a buzzer with Arduino and used the gas sensor reading to control it. A threshold value was selected in the program. Whenever the measured gas value became higher than the threshold, Arduino activated the buzzer automatically. If the gas value remained below the threshold, the buzzer stayed off.

After uploading the program, I tested the circuit by exposing the sensor to gas. The buzzer produced sound only when the gas level exceeded the selected threshold, indicating that the condition was working properly.

### Code for Problem 2:

```
const int gasPin = A0;
const int buzzer = 8;
int threshold = 100;

void setup()
{
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int gasValue = analogRead(gasPin);
  Serial.print("Gas Value: ");
  Serial.print(gasValue);
  if (gasValue > threshold)
  {
    digitalWrite(buzzer, HIGH);
    Serial.println("  --> Buzzer ON");
  }
  else
  {
    digitalWrite(buzzer, LOW);
    Serial.println("  --> Buzzer OFF");
  }
  delay(500);
}
```

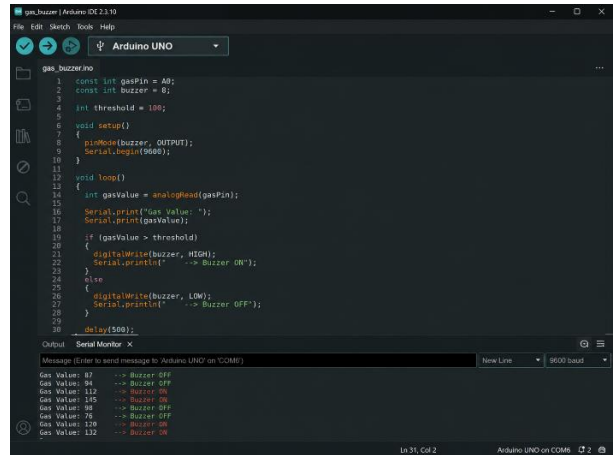
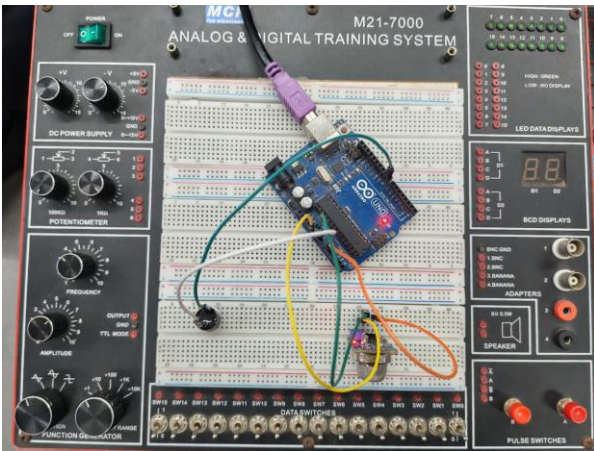


Figure: Problem 2

## Problem 3: Display "Gas is detected" on the OLED Screen

### Solution:

In this task, the OLED display was used to show a warning message whenever gas was detected. Arduino continuously monitored the sensor value. If the measured value crossed the predefined limit, the message **"Gas is detected"** appeared on the OLED display. Otherwise, the display continued showing the normal gas sensor reading.

After testing the circuit, I found that the warning message appeared immediately after gas was detected and disappeared when the gas level returned to a safe range.

### Code for Problem 3:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
const int gasPin = A0;
int threshold = 100;

void setup()
{
  Serial.begin(9600);
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
  {
    Serial.println("OLED Failed");
    while(true);
  }
}
```

```

}
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(WHITE);
display.setCursor(0,0);
display.println("Gas Detector");
display.display();
delay(2000);
}

void loop()
{
    int gasValue = analogRead(gasPin);
    Serial.print("Gas Value: ");
    Serial.println(gasValue);
    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(0,0);
    display.print("Gas Value:");
    display.println(gasValue);
    if(gasValue > threshold)
    {
        display.setCursor(0,30);
        display.println("Gas is detected");
    }
    else
    {
        display.setCursor(0,30);
        display.println("No Gas");
    }
    display.display();
    delay(500);
}

```

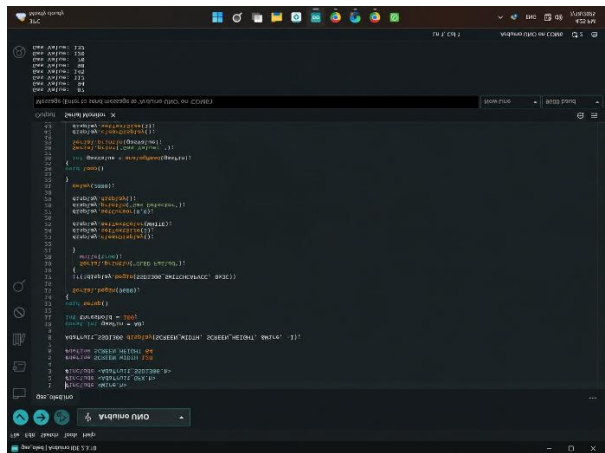
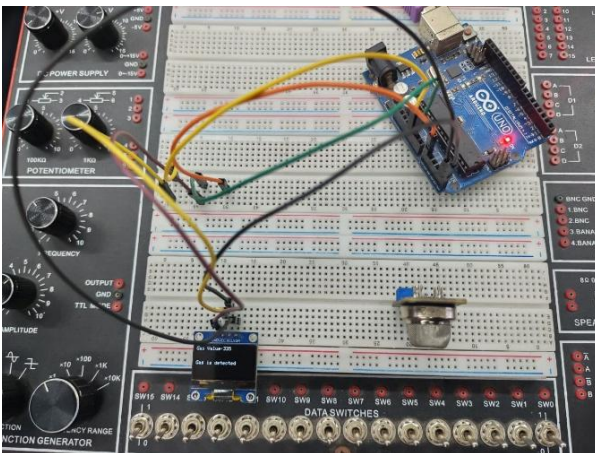


Figure: Problem 3

## Problem 5: Turn ON an LED When Gas is Detected

### Solution:

In this task, I connected an LED with Arduino Uno to provide a visual indication whenever gas was detected. Arduino continuously monitored the value received from the MQ-2 gas sensor. When the sensor value became higher than the predefined threshold, Arduino turned the LED on automatically. If the gas level returned to a safe range, the LED was turned off again.

After uploading the program, I tested the circuit by exposing the sensor to gas. The LED turned on immediately when gas was detected and turned off after the gas concentration decreased. The output matched the expected behavior and confirmed that the circuit was working correctly.

### Code for Problem 5:

```
const int gasPin = A0;
const int ledPin = 8;
int threshold = 100;

void setup()
{
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
  Serial.begin(9600);
  Serial.println("Gas Detection System Started...");
}
void loop()
{
  int gasValue = analogRead(gasPin);
  Serial.print("Gas Value: ");
  Serial.print(gasValue);
  if (gasValue > threshold)
  {
    digitalWrite(ledPin, HIGH);
    Serial.println(" --> Gas Detected! LED ON");
  }
  else
  {
    digitalWrite(ledPin, LOW);
    Serial.println(" --> Safe. LED OFF");
  }
  delay(500);
}
```



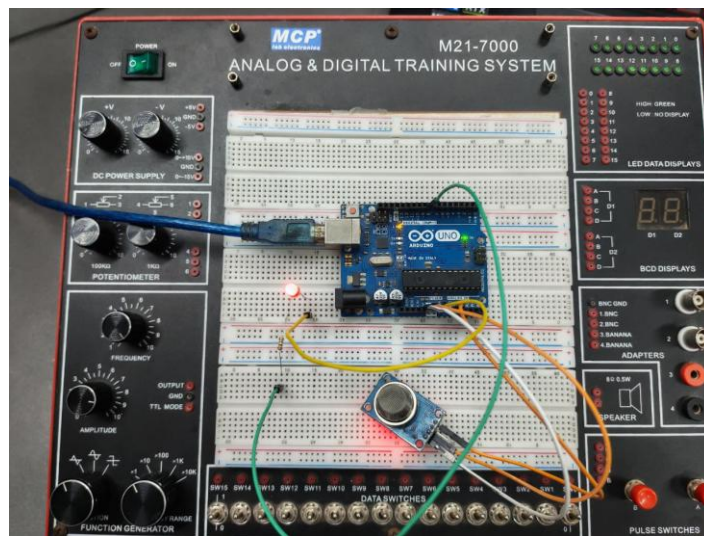
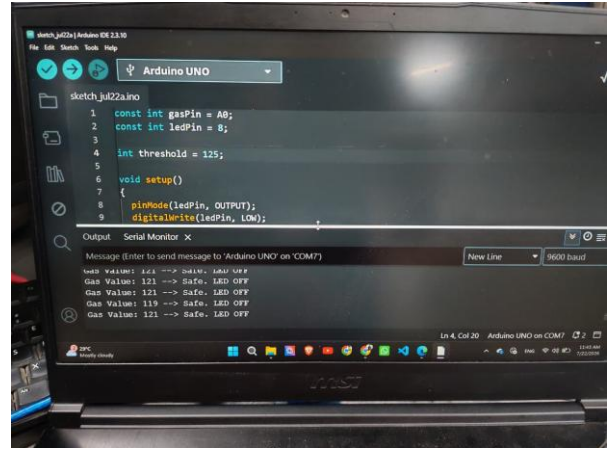
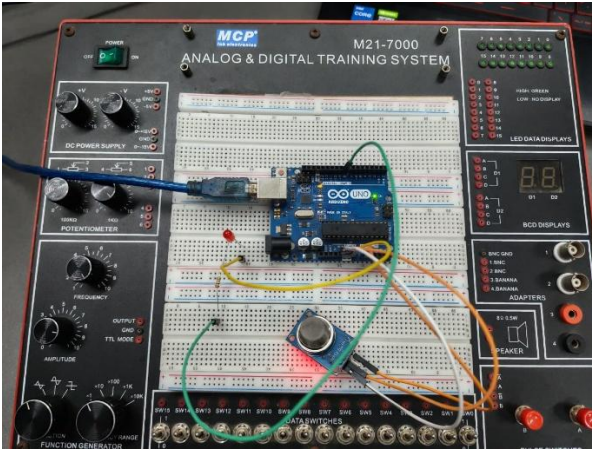


Figure: Problem 5

## Discussion:

In this experiment, I successfully interfaced the MQ-2 gas sensor with Arduino Uno and completed five different tasks using actual hardware components. First, I displayed the gas sensor reading on an OLED display. Next, I controlled a buzzer based on the measured gas concentration. I then displayed the message **"Gas is detected"** on the OLED whenever the gas level exceeded the selected threshold. After that, I implemented a simple gas alarm using the buzzer. Finally, I used an LED as a visual indicator that turned on automatically when gas was detected.

This experiment helped me understand how Arduino reads sensor data and controls different output devices based on programmed conditions. It also gave me practical experience in connecting hardware components and building a simple gas detection and warning system.